Check for updates

RESEARCH ARTICLE

REVISED A cross-sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka

[version 2; peer review: 2 approved]

Pracheth Raghuveer[1], Ravjot Bhatia[2], Rohith Motappa[2], Malavika Sachith[3]

[1]Epidemiology, National Institute of Mental Health and Neurosciences, India, Bangalore, Karnataka, 560029, India
[2]Community Medicine, Kasturba Medical College Hospital, Mangalore, Karnataka, 575001, India
[3]Community Medicine, P K DAS Institute of Medical Sciences, Ottapalam, Kerala, 679522, India

v2 **First published:** 21 Dec 2022, **11**:1546
https://doi.org/10.12688/f1000research.128273.1
Latest published: 04 Jul 2023, **11**:1546
https://doi.org/10.12688/f1000research.128273.2

Abstract

Background: Spirituality and spiritual health are an integral component of an individual's health and wellbeing. Among medical undergraduates and professionals, spiritual health has positive effects on the individual by decreasing burnout, psychological distress, and rates of substance abuse. Spiritual health is also correlated with increased satisfaction and meaning in life. Spiritual health also influences future patient care, builds patient- provider communication channels, and improves patient outcomes.

Methods: A cross-sectional study was conducted to assess the spiritual health levels of medical undergraduates in a medical college, in Karnataka, India, and to identify the factors associated with it. Using a pre-designed, validated semi-structured questionnaire, 600 medical undergraduates were approached and provided the forms. Spiritual health was assessed across the three domains of self-development, self-actualization, and self-realization using the Spiritual Health Assessment Scale (SHAS).

Results: A total of 436 medical undergraduates participated. Majority (67.7%) of participants were girls. Most (62%) were found to have fair spiritual health with a large portion of the rest (36%) having good spiritual health and 2% had poor spiritual health. A significant association was found between spiritual health and the father's (p=0.005) and mother's (p=0.012) education levels. Spiritual health was also found to be associated with living in a nuclear family (p=0.04).

Conclusions: Majority of the medical undergraduates had fair spiritual health. Parents' education levels and nuclear family were significantly associated with spiritual health.

Keywords

Spirituality, Medical Students, Dakshina Kannada

Open Peer Review

Approval Status ✔ ✔

	1	2
version 2 (revision) 04 Jul 2023	✔ view	✔ view
version 1 21 Dec 2022	? view	? view

1. **Vedalaveni Chowdappa Suresh**, Akash Institute of Medical Sciences and Research Centre, Bangalore, India

2. **Chonnakarn Jatchavala**, Prince of Songkla University, Songkhla, Thailand

Any reports and responses or comments on the article can be found at the end of the article.

This article is included in the Manipal Academy of Higher Education gateway.

Corresponding author: Rohith Motappa (drrohith88@gmail.com)

Author roles: Raghuveer P: Conceptualization, Formal Analysis, Methodology, Project Administration, Supervision, Writing – Original Draft Preparation; **Bhatia R**: Resources, Validation, Writing – Original Draft Preparation, Writing – Review & Editing; **Motappa R**: Conceptualization, Methodology, Validation, Writing – Original Draft Preparation; **Sachith M**: Data Curation, Formal Analysis, Methodology, Software, Writing – Original Draft Preparation, Writing – Review & Editing

Competing interests: No competing interests were disclosed.

Grant information: The author(s) declared that no grants were involved in supporting this work.

Copyright: © 2023 Raghuveer P *et al*. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

How to cite this article: Raghuveer P, Bhatia R, Motappa R and Sachith M. **A cross-sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka [version 2; peer review: 2 approved]** F1000Research 2023, **11**:1546 https://doi.org/10.12688/f1000research.128273.2

First published: 21 Dec 2022, **11**:1546 https://doi.org/10.12688/f1000research.128273.1

REVISED **Amendments from Version 1**

Changes made from Version 1 to Version 2 as per reviewers' comments:

1. We have further improvised the conclusion section of the abstract.
2. We have removed the keyword Dakshina Kannada and replaced it with Health Assessment. Kindly find the updated change in Line 2.
3. We have revised the definition in line 7 and 8.
4. We have modified the sentence to relationships between spiritual health and finding satisfaction and meaning in life (line 18-20).
5. We had included 83 final year students in the methodology. Changes done in line 51, page.
6. We have deleted the sentences pertaining to ethics committee clearance and permission form principal in lines 48, 70 and 71.
7. Variables i.e., gender, religion, year of study, occupation did not show any significant correlation with the spirituality. Please find the change in line 100.
8. Modified conclusion and made it specific to objectives in lines 174-176.
9. Lines 146-151 changed to include the study recommendations.

Any further responses from the reviewers can be found at the end of the article

Introduction

Spirituality, in its purest form, has been a vital aspect of an individual's health since time immemorial. Health, often understood as a singular entity, is made up of multiple dimensions, with the WHO defining it as "Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity."[1] While the three classical dimensions of health are rather extensive, there exists a fourth dimension, namely spiritual health.[2,3] Spiritual health, as defined by Dhar *et al.*, is "a state of being where an individual is able to deal with day-to-day life in a manner which leads to the realization of one's full potential; meaning and purpose of life; and happiness from within." Under this broad canopy of spiritual health, three aspects have been suggested which are, "Realization of one's Full Potential"; "Meaning and Purpose of Life"; and "Happiness From Within".[3]

Over the last few decades, researchers have explored the relationships that exist between spiritual health and finding satisfaction and meaning in life.[4] Spirituality has also been associated with a decreased risk for burnout among physicians and trainees.[5] A similar correlation has been observed among medical students, with higher levels of spirituality being associated with lower levels of burnout and psychological distress, with spirituality possibly acting as a protective factor.[6] Few studies have also found an inverse correlation between substance abuse and spirituality in college and medical students, with higher levels of spirituality correlating with lower rates of substance abuse as well as rates of binge drinking.[7,8]

Beyond the overall personal wellness of the students, spiritual health also affects physicians and the care and treatment provided by them. In an article by Isaac *et al.*, spiritual health was found to be related to the interpretation of illnesses by patients. They also identified spirituality to have a mixed effect on health behaviour changes, acting both as a positive as well as a negative factor. Spirituality may also act as a bridge, furthering the patient – provider relationship and improving healthcare outcomes for all.[9] Geriatric care and palliative care may especially benefit from a physician's spiritual health, by helping them explore and build a healthy connection with the patient, encouraging open and effective communication. Communication and openness play a vital role in patient centred care, and boost health outcomes.[10]

The importance of spirituality for the wellness of a medical student as well as its impact on their care taking abilities cannot be understated. High level of spiritual health will improve the overall wellness of doctors, as well as the quality of care provided by them. While a few studies have been conducted to assess factors associated with spirituality, they are far and few in between, and even fewer exist for this region, and for this demographic that may be disproportionately benefited from spiritual wellness. Therefore, we carried out this study to assess the spiritual health of undergraduate medical students and the factors associated with it.

Methods

The study was conducted on the undergraduate medical students of 4 batches (150 × 4) i.e. 600 students. Complete enumeration was done to include all medical students of Yenepoya Medical College from First to Final year of Bachelor of Medicine and Bachelor of Surgery (MBBS) course. Each batch has 150 students. The study was carried out between December 2020 to January 2021. Those unwilling to give consent, those who did not own a phone or did not understand English were excluded from the study.

A pre-designed, validated, semi-structured proforma was prepared. This proforma was incorporated into a Google form. This was sent to the class representatives of each batch requesting them to distribute among their batchmates. The form

had a participant information sheet and informed consent followed by the pre-designed questionnaire. The questionnaire included demographic variables like age, sex, religion, year of study. Spiritual Health Assessment Scale (SHAS) was be used to assess the three domains of spiritual health like self-development, self-actualization and self-realization.[11] It involves 21 questions, individual scores in SHAS ranges from 21 to 105 and the level of spiritual health can be further graded as poor (21 to 49), fair (50 to 77) and good (78 to 105). Questions pertaining to the factors associated with it were also included.

The Google form responses were downloaded in a comma separated value format and the data captured was further cleaned. Descriptive statistics like mean, frequency and proportion were applied. Chi-square test were used to assess the association between level of spiritual health and factors associated with it.

Clearance from the Institutional Ethics Committee was obtained. Written informed consent was obtained from all the study participants. Strict confidentiality of the information collected was maintained. All the data was kept confidential.

Results

In the study, a total of 600 students were approached for the study, out of which 436 students responded. The response rate was 72.7%. The sociodemographic data is shown in Table 1. Most of the students (69.7%), belonged to female gender. About half the students (49.1%) students were Muslims while Hindus constituted 41.3% of the participants. Participation from the final year students was the least at 19%. A large majority (85.8%) were in a nuclear family while 14.2% were part of a joint family.

Table 1. Sociodemographic features of the medical students (N=436).

	N	%
Gender		
Male	132	30.3
Female	304	69.7
Religion		
Hindu	180	41.3
Christian	38	8.7
Muslim	214	49.1
Others	4	0.9
Year of study		
First year	107	24.5
Second year	138	31.7
Third year	108	24.8
Final year	83	19
Type of family		
Nuclear family	374	85.8
Joint family	62	14.2
Father's education		
Uneducated	3	0.7
Schooling	78	17.9
Graduate	203	46.6
Post graduate and above	152	34.9
Mother's education		
Uneducated	5	1.1
Schooling	116	26.6
Graduate	207	47.5
Post graduate and above	108	24.8

Table 1. *Continued*

	N	%
Father's occupation		
Government	87	20
Private	154	35.3
Self employed	162	37.2
Retired	31	7.1
Unemployed	2	0.5
Mother's occupation		
Government	72	16.5
Private	60	13.8
Self employed	67	15.4
Retired	8	1.8
Unemployed	229	52.5

Fathers' education was analysed, and it was found that 46.6% were graduates, 34.9% were postgraduates and above. On the other hand, 47.5% mothers were graduates while 24.8% were postgraduates. Majority of the fathers (37.2%) were self-employed. More than half (52.5%) of the mothers were homemakers.

Table 2 shows the range of spiritual health obtained by the students. Self-development varied from 11 to 34 with a median of 26. Range of self-actualisation score was 7 to 35 and median was 26. Final spiritual health score varied from 40 to 103 with a median of 74.

It was found that a graphical representation of the findings is displayed in Figure 1.

Table 2. The range of spiritual health scores obtained by the medical students.

Spiritual health variables	Median (IQR)	Range of scores found in the survey	
		Maximum value	Minimum value
Self-development	26 (4)	34	11
Self-actualisation	26 (5)	35	7
Self-realisation	23 (6)	35	7
Final spiritual health score	74 (14)	103	40

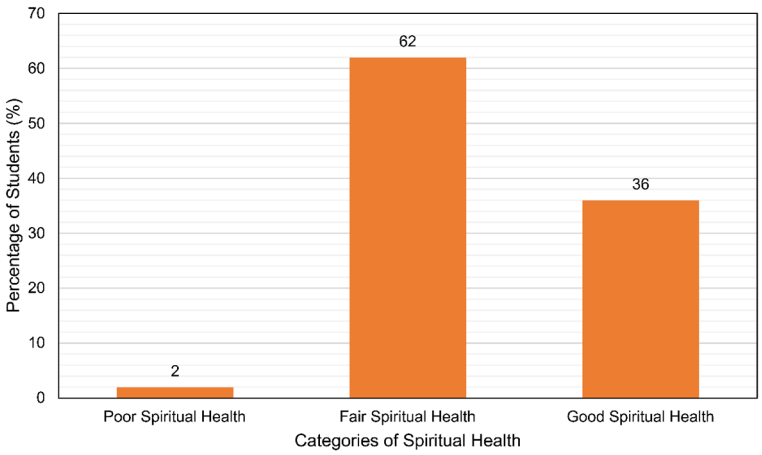

Figure 1. The spiritual health levels of the undergraduate medical students.

Table 3. Association of various demographic factors and spiritual health (N=436).

S. No	Variable		Categories of spiritual health			p- value*
			Poor	Fair	Good	
1.	Gender	Male (n=132)	4 (3.03%)	74 (56.06%)	54 (40.90%)	0.2
		Female (n=304)	4 (1.31%)	195 (64.14%)	105 (34.53%)	
2.	Religion	Hindu (n=180)	3 (1.66%)	123 (68.33%)	54 (30.0%)	0.1
		Christian (n=38)	2 (5.26%)	24 (63.15%)	12 (31.57%)	
		Muslim (n=214)	3 (1.40%)	119 (55.60%)	92 (42.99%)	
		Others (n=4)	0 (0)	3 (75.0%)	1 (25.0%)	
3.	Year of study	First year (n=107)	1 (0.93%)	70 (65.42%)	36 (33.64%)	0.56
		Second year (n=138)	2 (1.44%)	83 (60.14%)	53 (38.40%)	
		Third year (n=108)	3 (2.77%)	71 (65.74%)	34 (31.48%)	
		Final year (n=83)	2 (2.40%)	45 (54.21%)	36(43.37%)	
4.	Type of family	Nuclear family (n=374)	8 (2.13%)	237 (63.36%)	129 (34.49%)	**0.04**#
		Joint family (n=62)	0 (0)	32 (51.61%)	30 (48.38%)	
5.	Father's education	Uneducated (n=3)	1 (33.33%)	1 (33.33%)	1 (33.33%)	**0.005**#
		Schooling (n=78)	1 (1.28%)	46 (58.97%)	31 (39.74%)	
		Graduate (n=203)	2 (0.98%)	128 (63.05%)	73 (35.96%)	
		Postgraduate and above (n=152)	4 (2.63%)	94 (61.84%)	54 (35.52%)	
6.	Mother's education	Uneducated (n=5)	1 (20.0%)	2 (40.0%)	2 (40.0%)	**0.012**#
		Schooling (n=116)	0 (0)	64 (55.17%)	52 (44.82%)	
		Graduate (n=207)	2 (0.96%)	137 (66.18%)	68 (32.85%)	
		Postgraduate and above (n=108)	5 (4.62%)	66 (61.11%)	37 (34.25%)	
7.	Father's occupation	Government (n=87)	2 (2.29%)	56 (64.36%)	29 (33.33%)	0.98
		Private (n=154)	2 (1.29%)	96 (62.33%)	56 (36.36%)	
		Self employed (n=162)	3 (1.85%)	96 (59.25%)	63 (38.88%)	
		Retired (n=31)	1 (3.22%)	20 (64.51%)	10 (32.25%)	
		Unemployed (n=2)	0 (0)	1 (50.0%)	1 (50.0%)	
8.	Mother's occupation	Government (n=72)	2 (2.77%)	50 (69.44%)	20 (27.77%)	0.50
		Private (n=60)	2 (3.33%)	39 (65.0%)	19 (31.66%)	
		Self employed (n=67)	2 (2.98%)	38 (56.71%)	27 (40.29%)	
		Retired (n=8)	0 (0)	4 (50.0%)	4 (50.0%)	
		Unemployed (n=229)	2 (0.87%)	138 (60.26%)	89 (38.86%)	

*Chi-square test.
#Statistically significant.

Using the chi-square test, association of various demographic factors with spiritual health was analysed and is shown in Table 3. Upon analysis, out of all the variables studied, three of them were found to display significant association. Type of family ($p<0.05$), father's education level and mother's education level were found to be significantly associated with spiritual health. On the other hand, the other studied variables i.e., gender, religion, year of study, occupation of father and mother did not show any significant correlation spiritual health.

Discussion

We studied a variety of demographic factors to identify the ones which share any correlation with spiritual health. A majority of the students were female. Most of the respondents were believers of Hinduism or Islam, with Christians filling in most of the remaining percentage. A roughly equal breakdown of responses were seen across the different years of study. Most of the respondents live in a nuclear family.

Out of all the variables assessed, fathers' and mother's education levels were found to be significantly associated with spiritual health. This result is similar to a study conducted by Kalpana *et al.*, among arts and science college students.[12] This could indicate and uncover the role a person's family background plays in the spiritual wellness of an individual. The adage "education begins at home", could very well be applied here. Parents and families are the first teachers of children. The foundational knowledge of a person is to a large extent influenced by the teachings of their parents and their family environment. Among other things, spiritual wellness and its associated techniques are often taught at home, with parents acting as the conduits of spirituality. Old beliefs, customs and rituals are passed down from generation to generation and play a vital role in the healthy exploration of spirituality.

Education and it's association with spiritual health should not be ignored. Teaching spiritual health components and techniques could be a possible way to counter and improve upon existing deficits in this aspect. The role of education should be considered and requires further exploration.

A significant association was noted between spirituality and nuclear families. This is unlike the findings of one study conducted by Ahangar *et al.*, which found a higher level of spirituality in adolescents living in joint families, compared to nuclear families.[13] This could possibly be due to the fact that the college is located in an urban area, which may have more nuclear families. Another factor is that due to the type of college, most of the students belong to higher socioeconomic stratas, which are often associated with independent nuclear family lifestyles.

In our study, 36% of the students had good spiritual health while 62% had fair spiritual health. 2% of the students had poor spiritual health. This proportion is somewhat similar to another study, which found a similar breakdown among dental college students, with the majority (74.55%) having fair spiritual health.[14] This could be attributed to a similar demographic of the respondents, and supports and strengthens the outcomes of this study.

A few studies have found gender to be a significant variable which we could not identify with significance.[14–16] The rest of the variables were not found to have any significant association.

The lack of a significant association between religion and spirituality could hint towards the general nature of spiritual health regardless of the conforming religion of the respondents. Common approaches could thus possibly be designed and implemented in order to boost spiritual health among the surveyed demographic.

Overall, few studies have been conducted in order to assess the variables associated with this often neglected domain of spiritual health and wellness. These variables must be targeted as early as possible to improve spiritual health. As mentioned previously, the effects of spiritual health reach far and beyond, especially for medical students and future physicians. Therefore, programs and approaches targeting spiritual health should be designed for which further research would be required in order to solidify the associations as well as identify other possible factors influencing and affecting spiritual health.

Strengths and limitations

Our study had several strengths. The large sample size afforded allows us a greater degree of reliability and the ability to generalize the results to other similar demographics. The design of this study also makes it easy to replicate and conduct further assessments across other demographics, as well as solidify the associations.

The limitations of this study include the fact that spiritual health has different definitions for different respondents, and thus the interpretation of the questions could vary from respondent to respondent, based on their own experiences with it. The second limitation is that this is a cross sectional study, and thus provides possible correlations and associations only. Another limitation that arises is that Yenepoya Medical College, a prestigious medical college in Mangalore, India, is a private college. Private colleges usually have higher fee structures associated with them; thus, the students usually belong to a higher socioeconomic status than government colleges, which have a healthy mix. Therefore, the results, which while useful and suitable for this demographic, cannot be fully generalized to students in all kinds of colleges. Nevertheless, spiritual health, the focus of our study, could play an important role in the overall health of a medical student as well as the outcomes of patient-centred cared that may be provided by the student in future.

Conclusion

Majority of the undergraduate medical students were found to have fair spiritual health. We have analyzed a few variables that could have possibly been associated with spiritual health. We found that that parents' education levels and family type as the significant factors associated with spiritual well-being.

Data availability

Underlying data

Figshare: A cross sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka (Responses) (2).xlsx, https://doi.org/10.6084/m9.figshare.21509781.v1.[17]

Extended data

Figshare: A cross-sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka. Informed Consent form.docx, https://doi.org/10.6084/m9.figshare.21601029.v1.[18]

Figshare: A cross-sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka. Participant Information Sheet.docx, https://doi.org/10.6084/m9.figshare.21601023.[19]

Figshare: A cross-sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka. Questionnaire.docx, https://doi.org/10.6084/m9.figshare.21601005.v1.[20]

Data are available under the terms of the Creative Commons Attribution 4.0 International license (CC-BY 4.0).

References

1. Park K: *Park's Textbook of preventive and social medicine.* 25th ed. Jabalpur: Banarsidas Bhanot Publishers; 2019; pp. 13–16.
2. Ali SM: **Spiritual well-being: the fourth dimension of health.** *Indian J. Public Health.* 2012; **56**: 257–258.
PubMed Abstract | Publisher Full Text
3. Dhar N, Chaturvedi S, Nandan D: **Spiritual health scale 2011: defining and measuring 4 dimension of health.** *Indian J. Community Med. Off. Publ. Indian Assoc. Prev. Soc. Med.* 2011; **36**: 275–282.
Publisher Full Text
4. Deb S, Thomas S, Bose A, *et al.*: **Happiness, Meaning, and Satisfaction in Life as Perceived by Indian University Students and Their Association with Spirituality.** *J. Relig. Health.* 2020; **59**: 2469–2485.
PubMed Abstract | Publisher Full Text
5. Collier KM, James CA, Saint S, *et al.*: **The Role of Spirituality and Religion in Physician and Trainee Wellness.** *J. Gen. Intern. Med.* 2021; **36**: 3199–3201.
PubMed Abstract | Publisher Full Text | Free Full Text
6. Wachholtz A, Rogoff M: **The relationship between spirituality and burnout among medical students.** *J. Contemp. Med. Educ.* 2013; **1**: 83–91.
PubMed Abstract | Publisher Full Text | Free Full Text
7. Williams MK, Greene WM, Leitner SA, *et al.*: **Medical Student Spirituality and Substance Use.** *J. Addict. Med.* 2020; **14**: e316–e320.
PubMed Abstract | Publisher Full Text
8. Stewart C: **The influence of spirituality on substance use of college students.** *J. Drug Educ.* 2001; **31**: 343–351.
PubMed Abstract | Publisher Full Text
9. Isaac KS, Hay JL, Lubetkin EI: **Incorporating Spirituality in Primary Care.** *J. Relig. Health.* 2016; **55**: 1065–1077.
PubMed Abstract | Publisher Full Text | Free Full Text
10. Balducci L: **Geriatric Oncology, Spirituality, and Palliative Care.** *J. Pain Symptom Manag.* 2019; **57**: 171–175.
Publisher Full Text
11. Gaur KL, Sharma M: **Measuring Spiritual Health: Spiritual Health assessment Scale (SHAS).** *Int. J. Innov. Res. Dev.* 2014; **3**: 63–67.
12. Devi RK, Rajesh NV, Devi MA: **Study of Spiritual Intelligence and Adjustment Among Arts and Science College Students.** *J. Relig. Health.* 2017; **56**: 828–838.
PubMed Abstract | Publisher Full Text
13. Ahangar M, Khan M: **Spiritual Intelligence of Adolescents of Nuclear and Joint Families-A Comparative Study.** *Int. Res. J. Hum. Resour. Soc. Sci.* 2017; **4**: 303–313. ISSNO 2349-4085 ISSNP 2394-4218.
14. Dhama K , *et al.*: **An Insight into Spiritual Health and Coping Tactics among Dental Students; A Gain or Blight: A Cross-sectional Study.** *J. Clin. Diagn. Res. JCDR.* 2017; **11**: ZC33–ZC38.
15. Pant N, Srivastava SK: **The Impact of Spiritual Intelligence, Gender and Educational Background on Mental Health Among College Students.** *J. Relig. Health.* 2019; **58**: 87–108.
PubMed Abstract | Publisher Full Text
16. Deb S, McGirr K, Sun J: **Spirituality in Indian University Students and its Associations with Socioeconomic Status, Religious Background, Social Support, and Mental Health.** *J. Relig. Health.* 2016; **55**: 1623–1641.
PubMed Abstract | Publisher Full Text
17. Motappa R: A cross sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka. figshare. [Dataset]. 2022.
Publisher Full Text
18. Motappa R: A cross-sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka. Informed Consent form. docx. figshare Extended data. 2022.
Publisher Full Text
19. Motappa R: A cross-sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka. Participant Information Sheet.docx. figshare. Extended data. 2022.
Publisher Full Text
20. Motappa R: A cross-sectional study on assessment of spiritual health and its associated factors among undergraduate students in a medical college in Mangalore, Karnataka. Dataset. *Questionnaire. docx. figshare.* 2022.
Publisher Full Text

Open Peer Review

Current Peer Review Status: ✔ ✔

Version 2

Reviewer Report 17 July 2023

https://doi.org/10.5256/f1000research.148603.r183926

© **2023 Suresh V.** This is an open access peer review report distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

✔ **Vedalaveni Chowdappa Suresh**
Department of Psychiatry, Akash Institute of Medical Sciences and Research Centre, Bangalore, Karnataka, India

No further comments to make.

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Mental Health

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Reviewer Report 11 July 2023

https://doi.org/10.5256/f1000research.148603.r183925

© **2023 Jatchavala C.** This is an open access peer review report distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

✔ **Chonnakarn Jatchavala**
Department of Psychiatry, Prince of Songkla University, Songkhla, Thailand

I am very satisfied with the revision. Nothing is to be changed or further revised.

Competing Interests: No competing interests were disclosed.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Version 1

Reviewer Report 13 February 2023

https://doi.org/10.5256/f1000research.140845.r158713

© **2023 Jatchavala C.** This is an open access peer review report distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

?

Chonnakarn Jatchavala
[1] Department of Psychiatry, Prince of Songkla University, Songkhla, Thailand
[2] Department of Psychiatry, Prince of Songkla University, Songkhla, Thailand

Major points:

1. This study is relatively new, we are talking about spiritual health which is very controversial. And it is a broad range to discuss.

2. In the results section of the abstract, please add more findings relevant to the study objectives.

3. Discussing empirical implications in medical education, authors need to find other cultures' perspectives on spiritual health that decrease burnout/ mental health issues or increase positive factors in higher education.

4. Could you modify the conclusion and make it specific to the study objectives?

5. What are the recommendations based on the study findings? Please include them in the discussion.

Minor points:

1. Please give us the odds ratio for the significantly associated factor if possible

2. Table 3 is quite enough. Table 2 could be noted: under table 2

Is the work clearly and accurately presented and does it cite the current literature?
Yes

Is the study design appropriate and is the work technically sound?
Yes

Are sufficient details of methods and analysis provided to allow replication by others?
Partly

If applicable, is the statistical analysis and its interpretation appropriate?
Partly

Are all the source data underlying the results available to ensure full reproducibility?

Partly

Are the conclusions drawn adequately supported by the results?
Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Mental health, higher education

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 22 May 2023

Rohith Motappa

What are the recommendations based on the study findings? Please include them in the discussion
Response. Thank you for the comment. Recommendations have been included in the discussion section last paragraph, Page 11.

Rest all comments have been addressed in the modified article file

Competing Interests: None

Reviewer Report 20 January 2023

https://doi.org/10.5256/f1000research.140845.r158712

© 2023 Suresh V. This is an open access peer review report distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

?

Vedalaveni Chowdappa Suresh
[1] Department of Psychiatry, Akash Institute of Medical Sciences and Research Centre, Bangalore, Karnataka, India
[2] Department of Psychiatry, Akash Institute of Medical Sciences and Research Centre, Bangalore, Karnataka, India

Abstract:
Conclusion: can be improvised. As both results and conclusion depicts same content.

Keywords: can consider removing Dakshina Kannada.

Introduction

1. Definition of health [reference 1] need to be re-written.

2. Definition of Spiritual health defined twice [two different references]

3. 2nd paragraph: relationship between spiritual health and what?

Methodology:

1. Any specific reason to include only 1st-3rd year students?

2. Ethical committee clearance mentioned twice. Since it is from same institution, need not mention about permission from principal in methods.

3. Study proforma had participant information sheet with **consent form** as mentioned, any reason for obtaining written consent separately?

4. Why was the domains like socio-economic status, background [rural/semiurban/urban] was not considered.

Results:

Could mention the variables ie. gender, religion, year of study, occupation did not show any significant correlation with the spirituality.

Is the work clearly and accurately presented and does it cite the current literature?

Yes

Is the study design appropriate and is the work technically sound?

Yes

Are sufficient details of methods and analysis provided to allow replication by others?

Yes

If applicable, is the statistical analysis and its interpretation appropriate?

Yes

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Mental Health

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 22 May 2023

Rohith Motappa

The definition of health [reference 1] needs to be re-written
Response: The definition has been rewritten and incorporated

2nd paragraph: the relationship between spiritual health and what?
Response: Thanks we have included it on page 2 of the manuscript in the introduction section, 3rd paragraph

Ethical committee clearance mentioned twice. Since it is from same institution, need not mention about permission from principal
Response: Thanks we have made the changes as advised. Ethics Committee clearance is mentioned just once. Please find the changes in the methods section, first and third paragraphs.

Study proforma had participant information sheet with consent form as mentioned, any reason for obtaining written consent separately?
Response: Thanks for the comment. We have removed the written consent part in methods section.

Competing Interests: None

The benefits of publishing with F1000Research:

- Your article is published within days, with no editorial bias
- You can publish traditional articles, null/negative results, case reports, data notes and more
- The peer review process is transparent and collaborative
- Your article is indexed in PubMed after passing peer review
- Dedicated customer support at every stage

For pre-submission enquiries, contact research@f1000.com

F1000Research